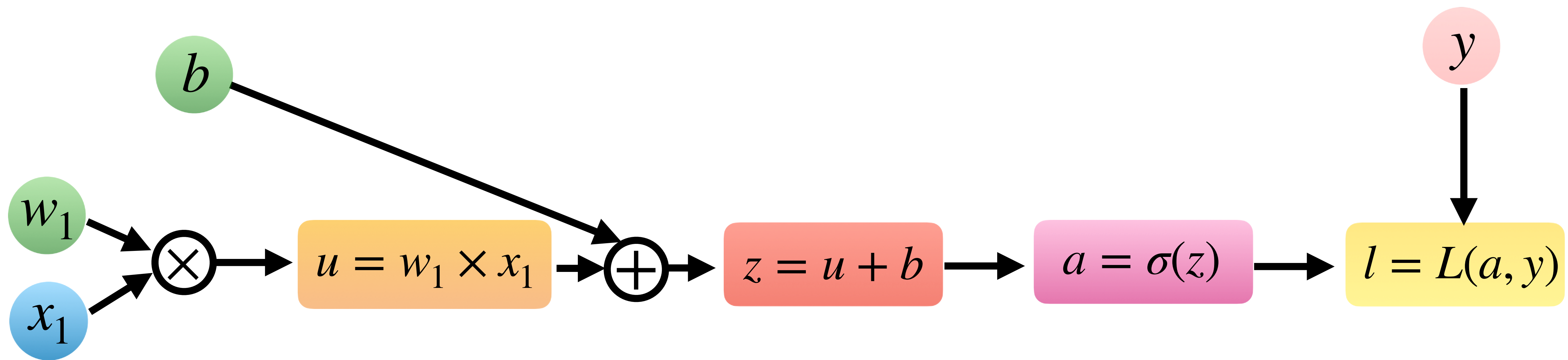


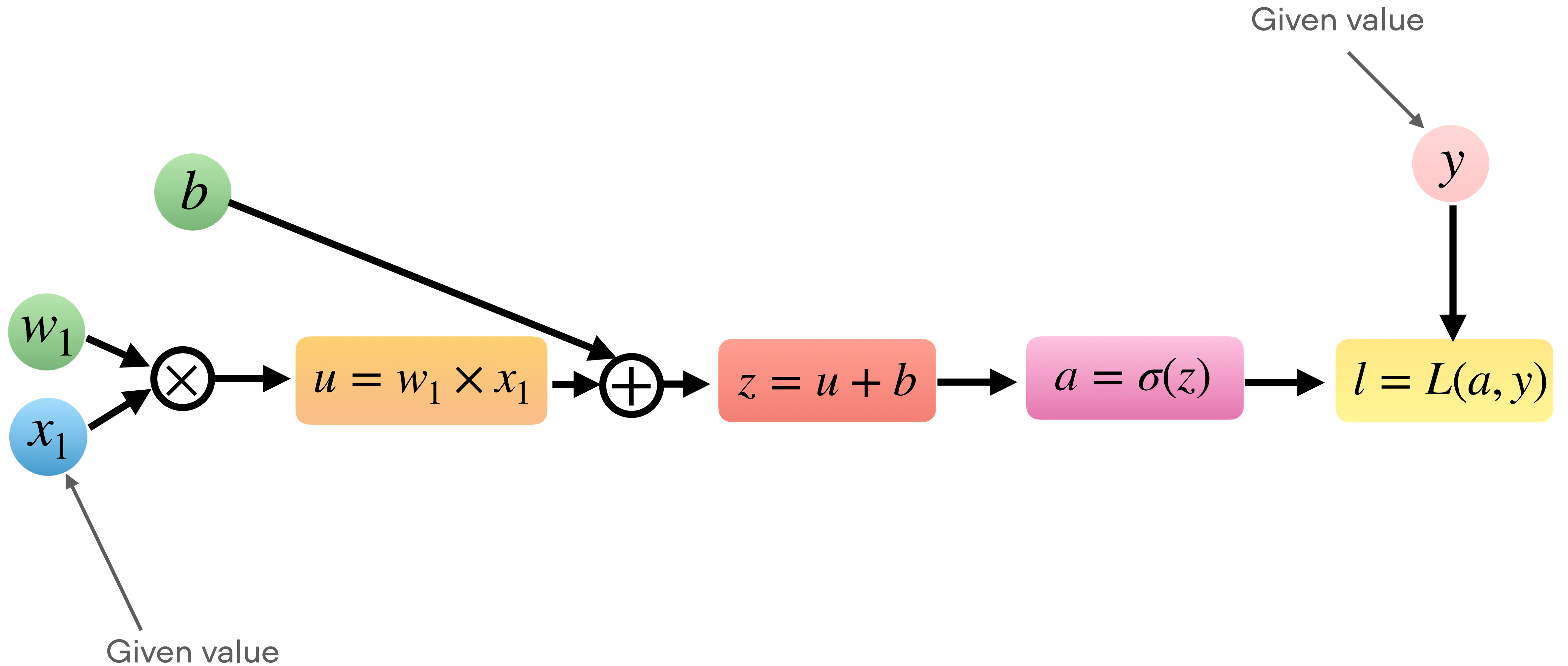
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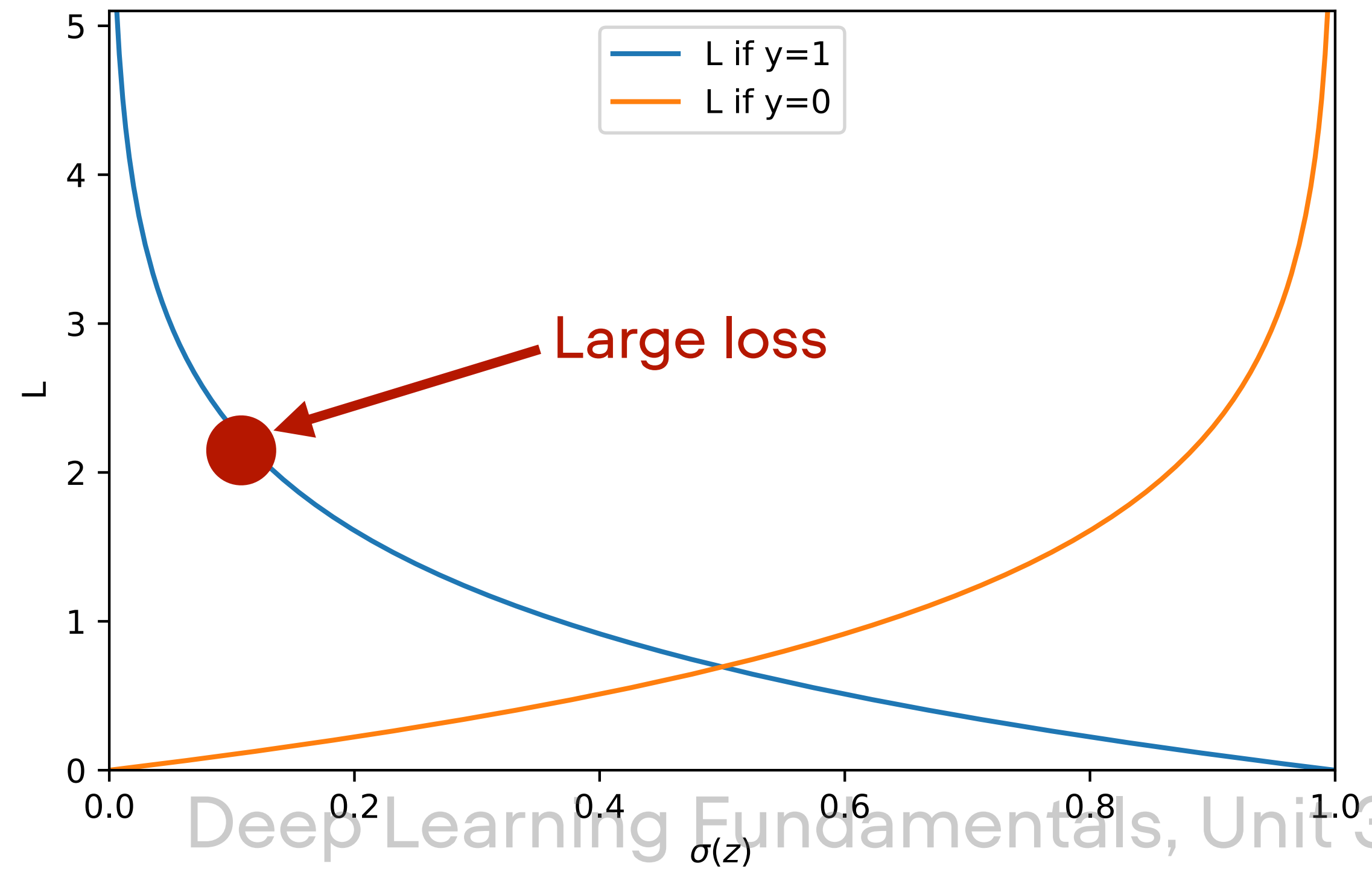
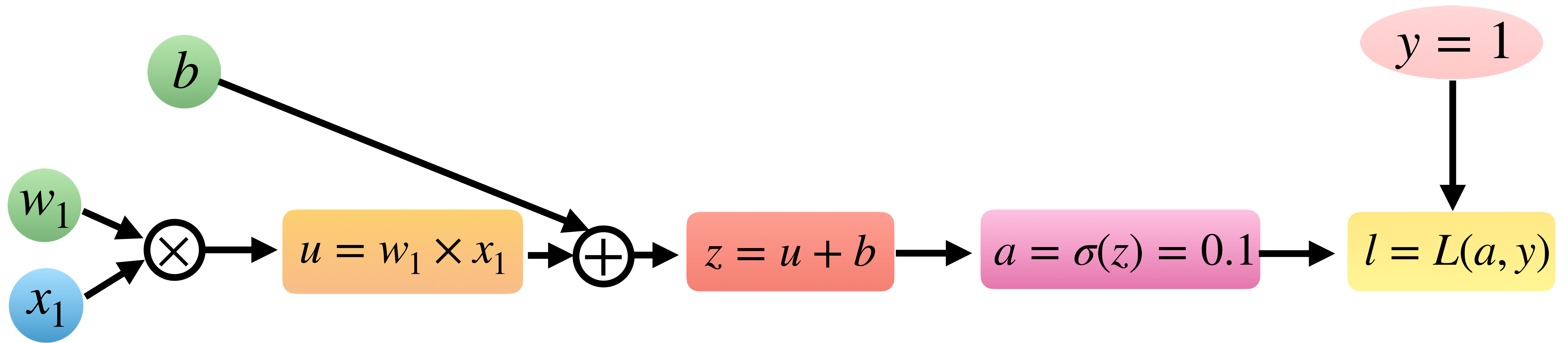
Model Training with Stochastic Gradient Descent

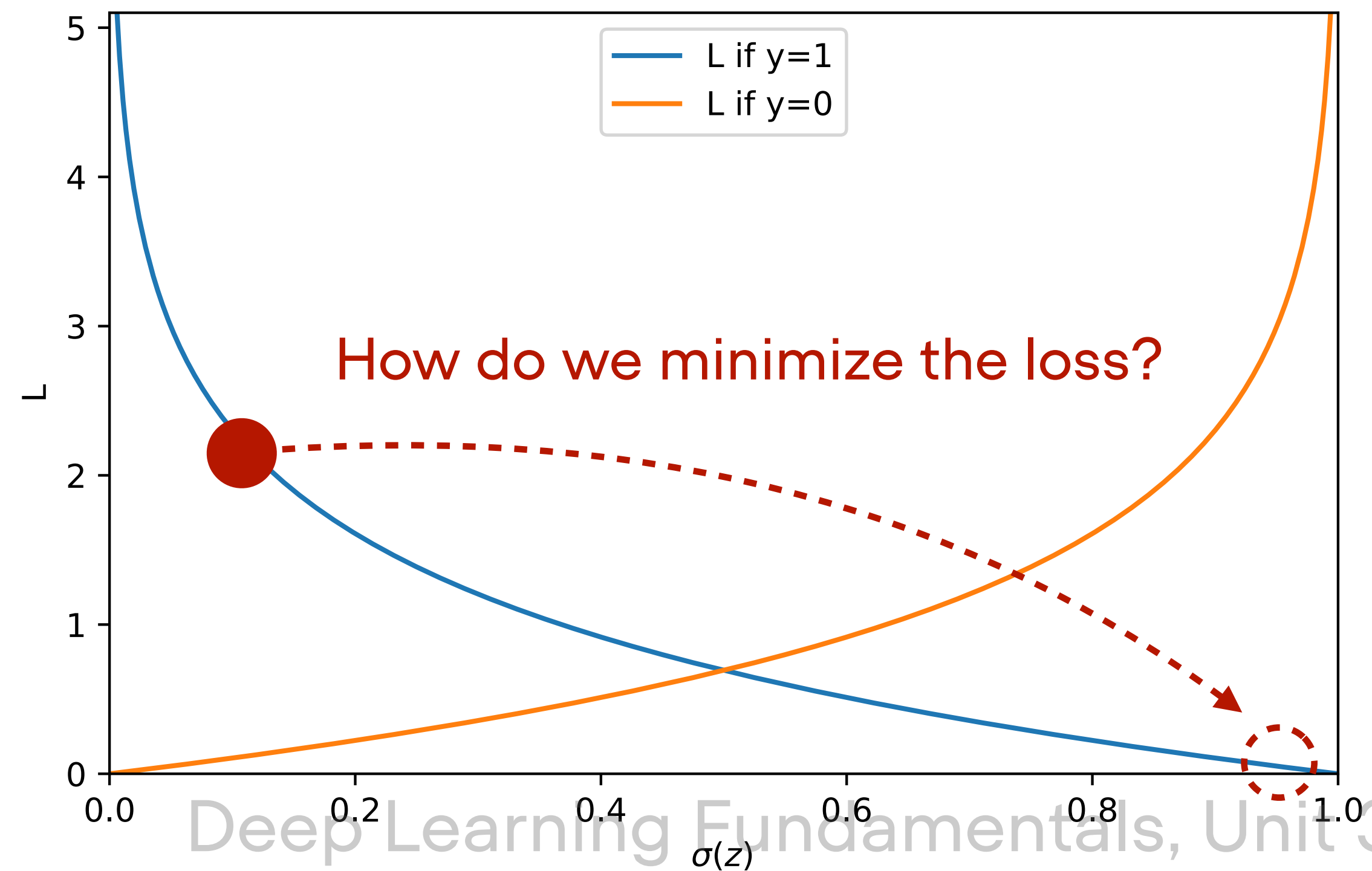
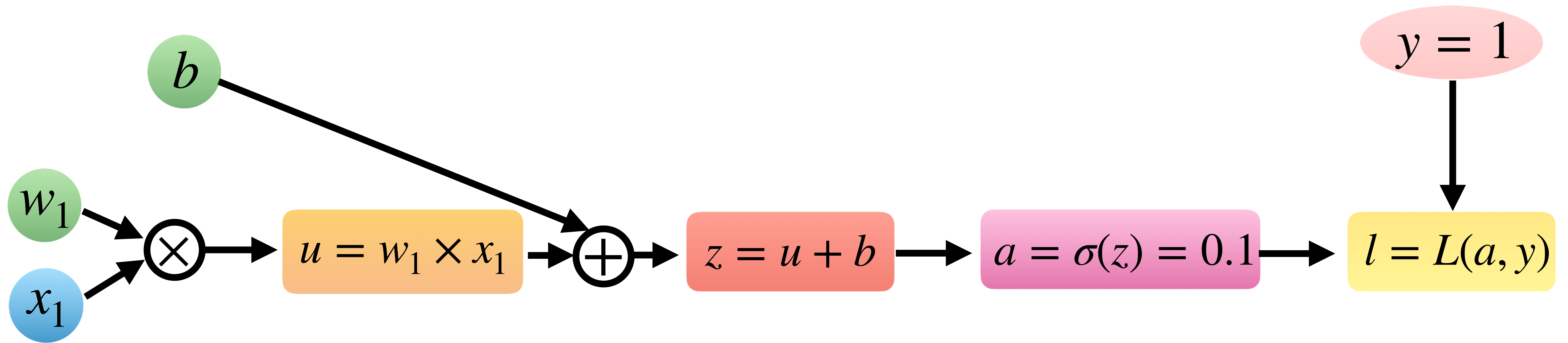
Part 1: Minimizing the Loss

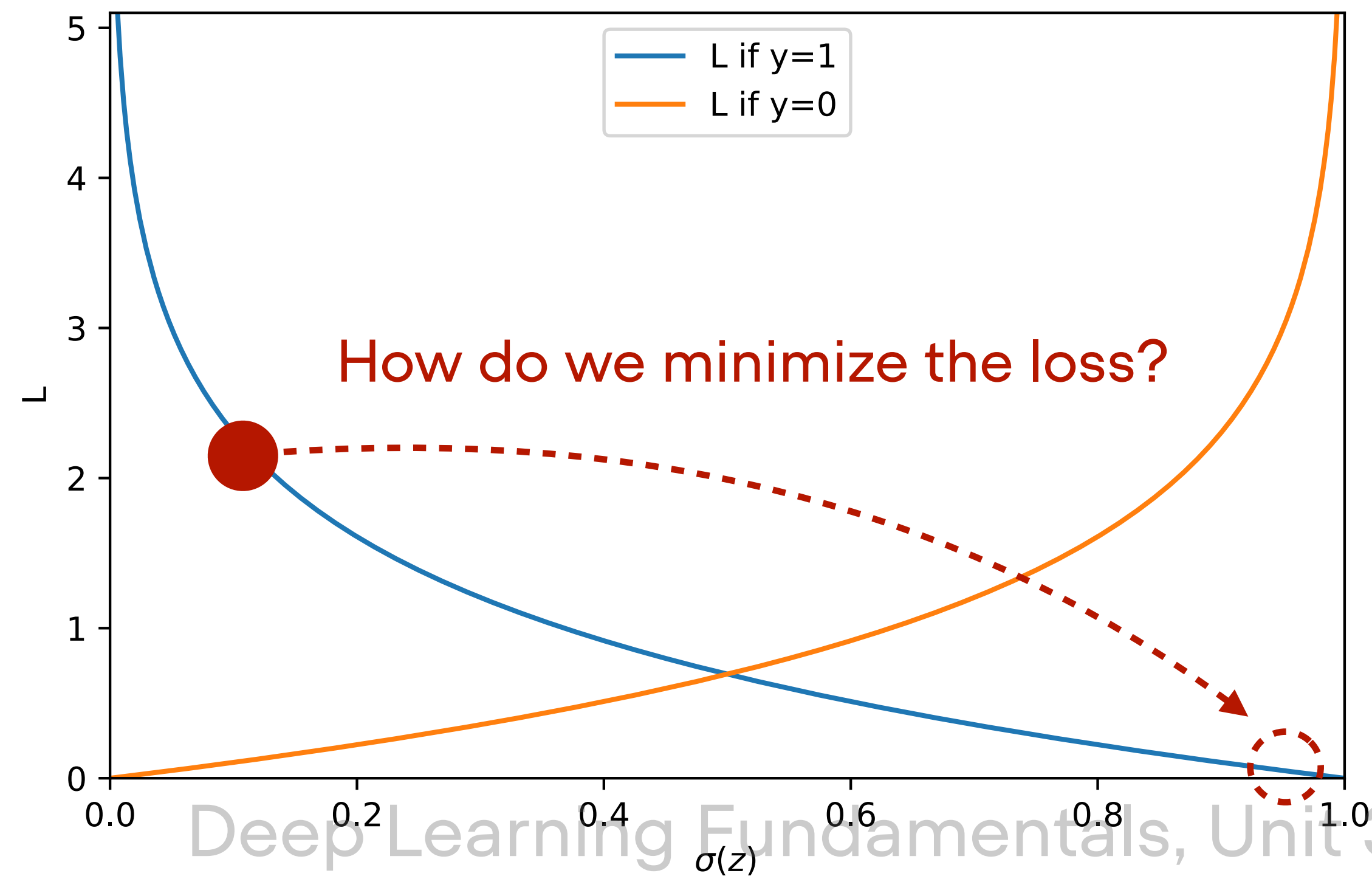
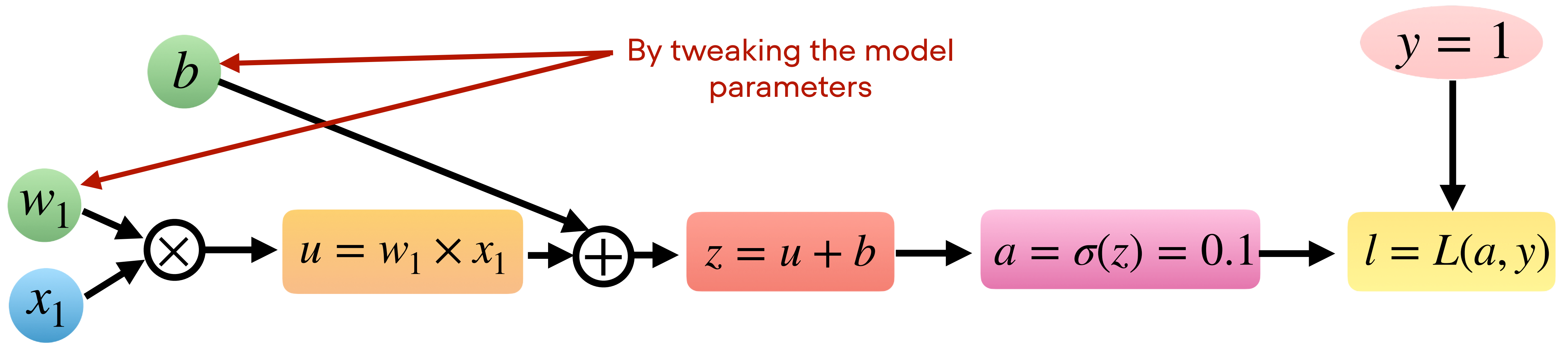
Sebastian Raschka and the Lightning AI Team

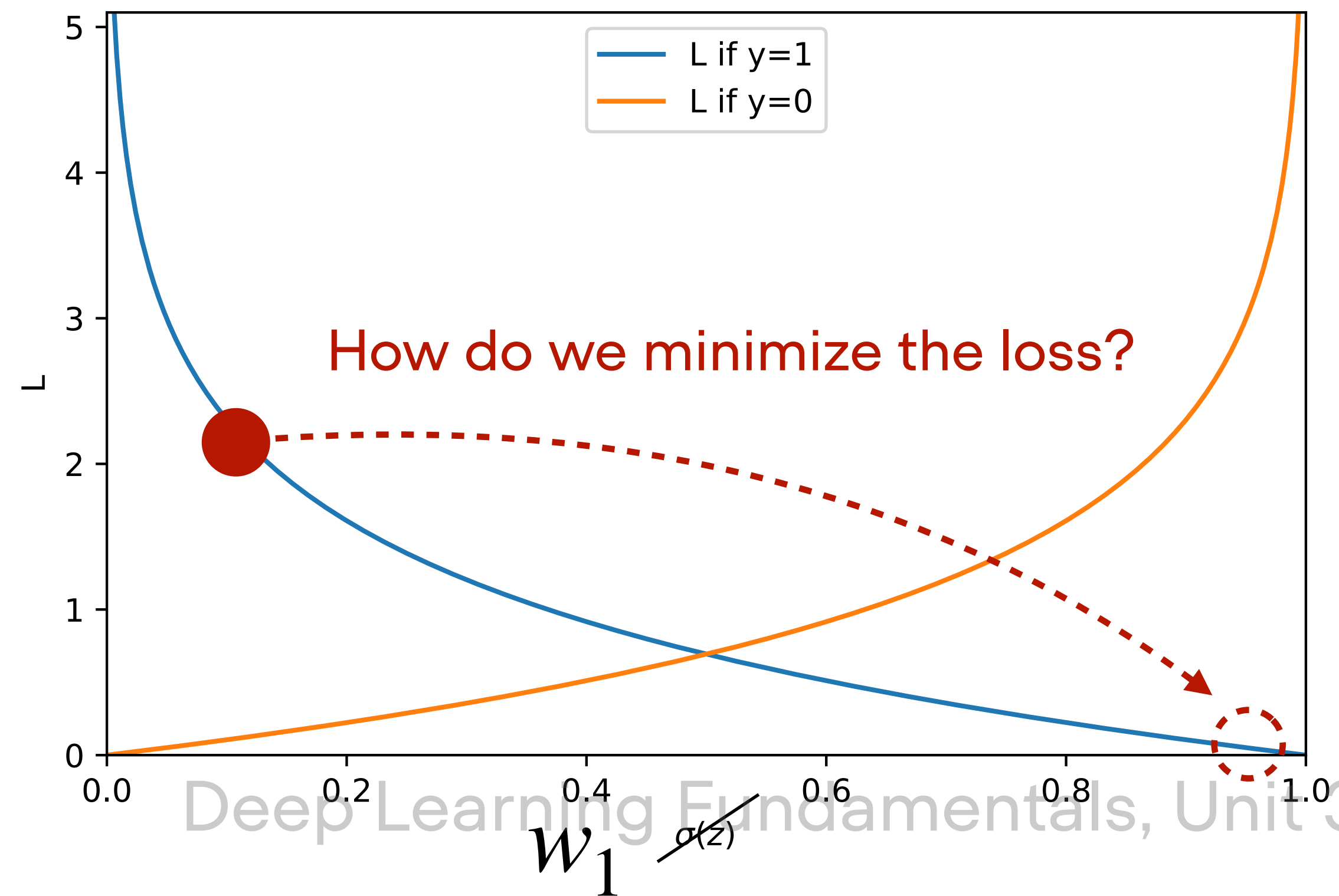
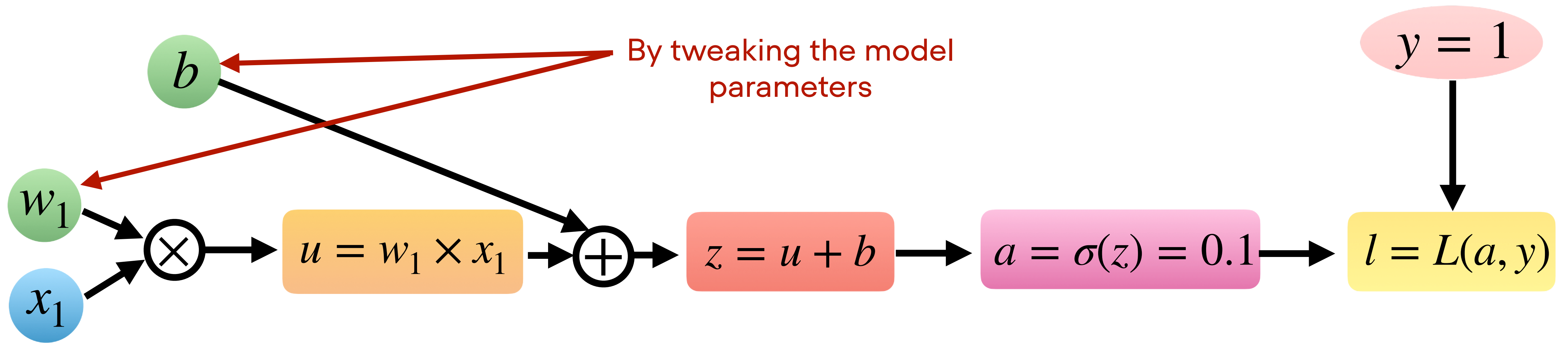




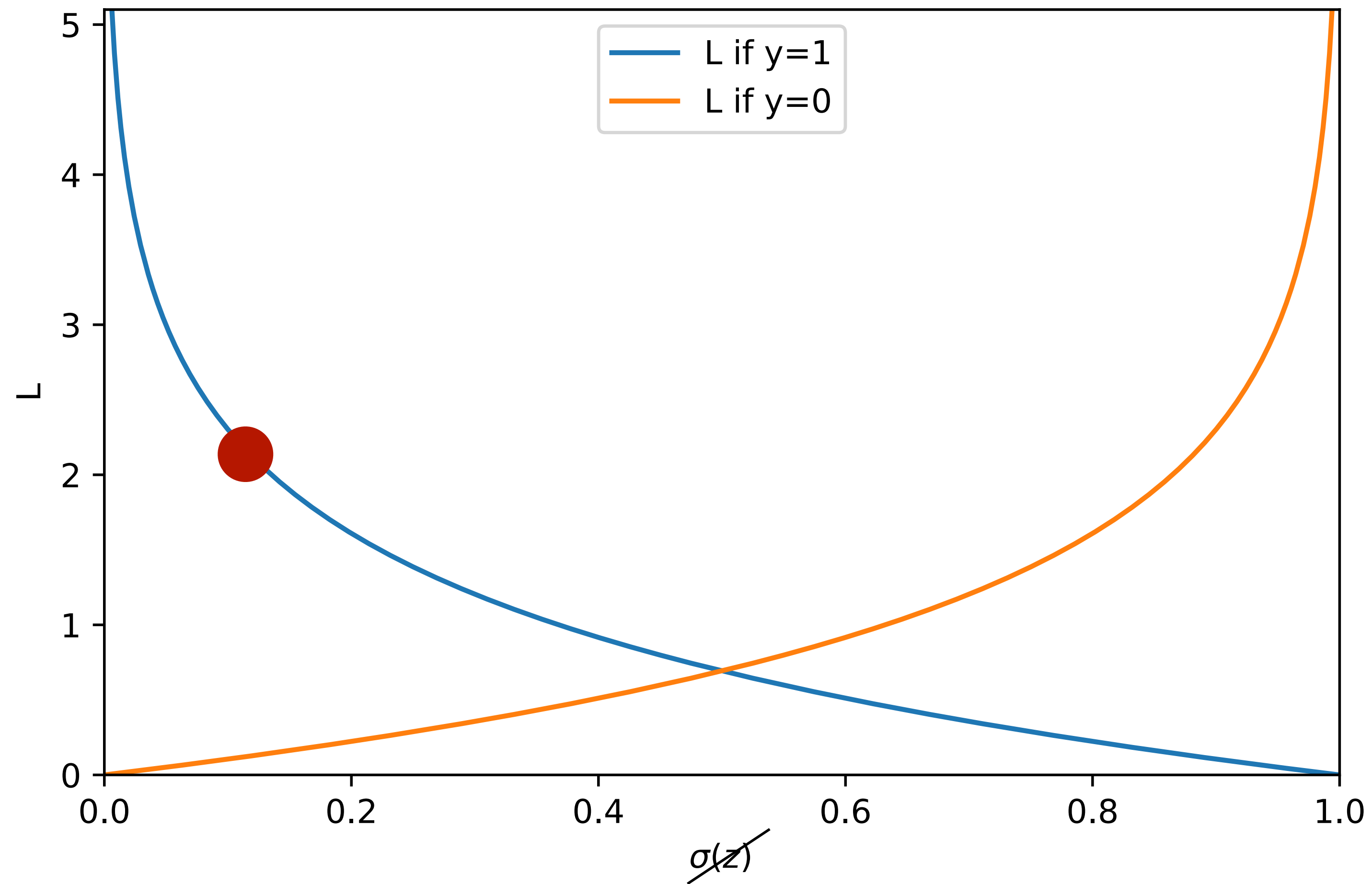




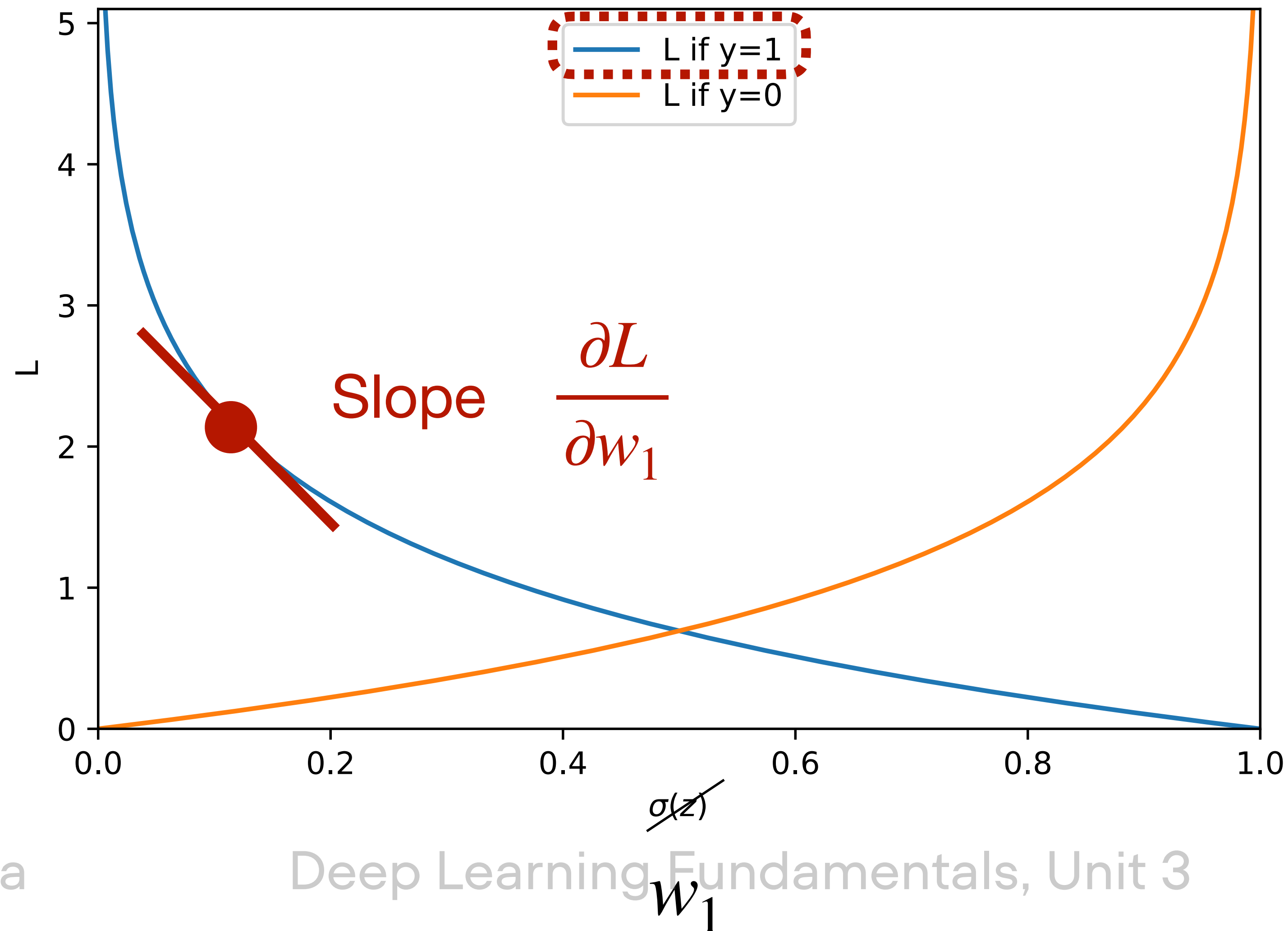




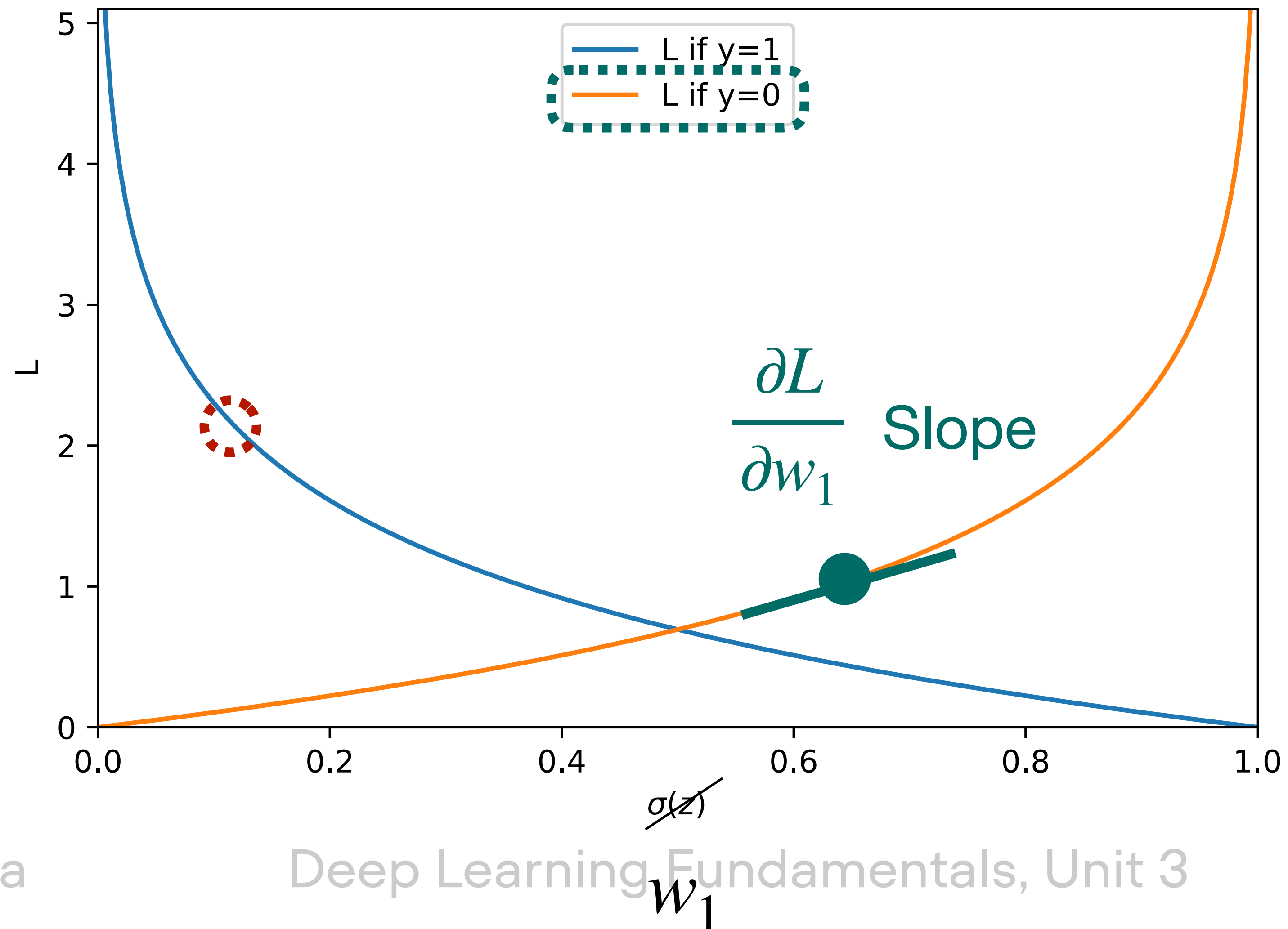
Changing parameters using the negative slope



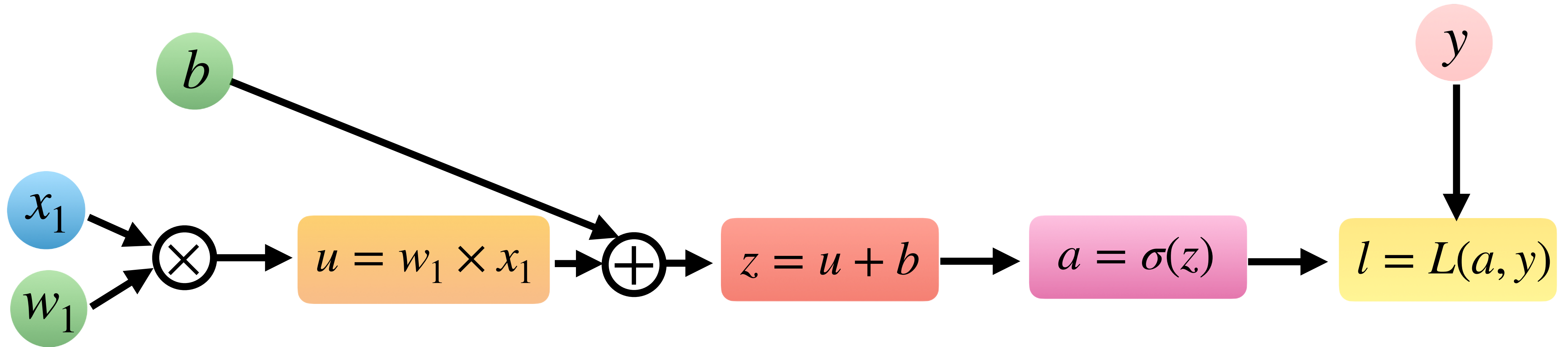
Changing parameters using the negative slope



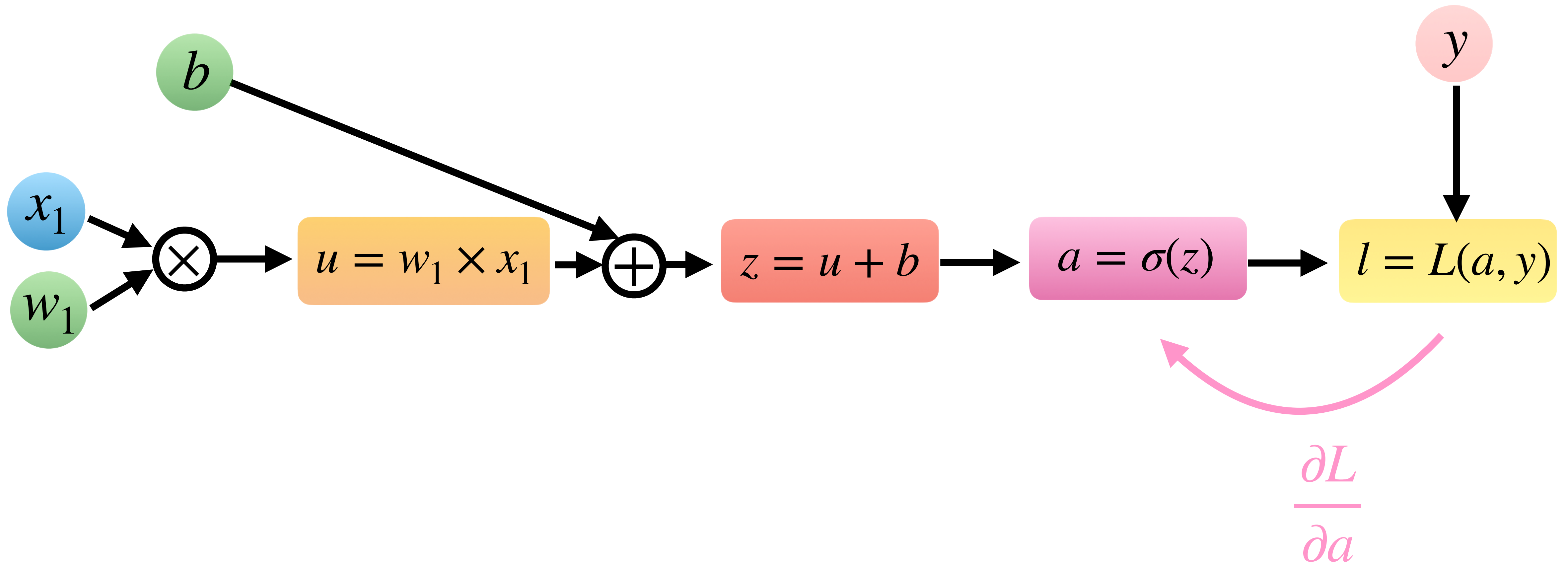
Changing parameters using the negative slope



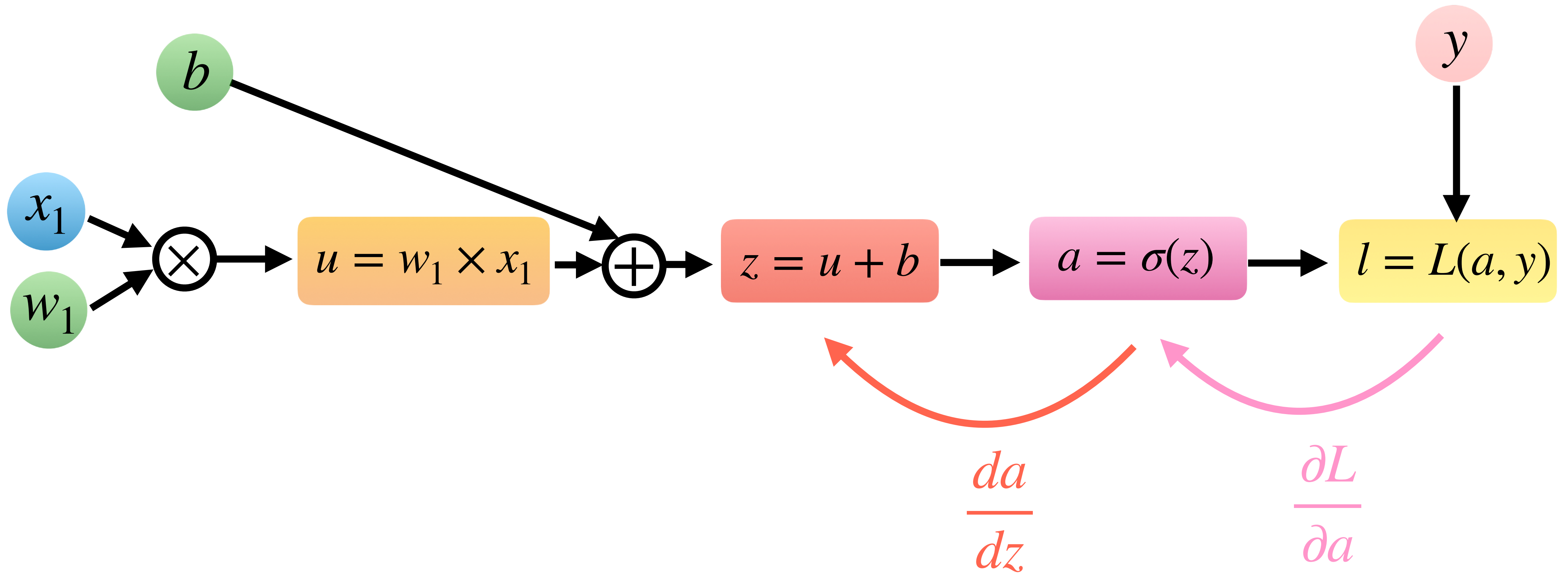
Chain Rule



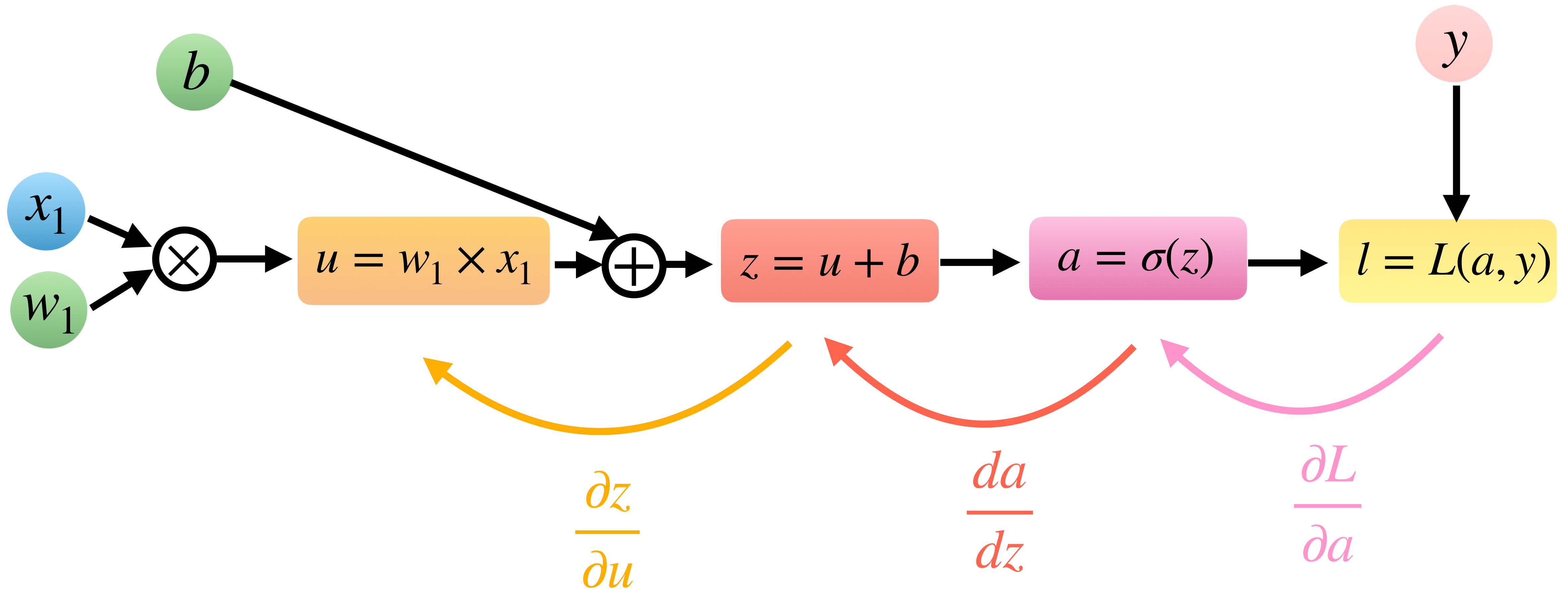
Chain Rule



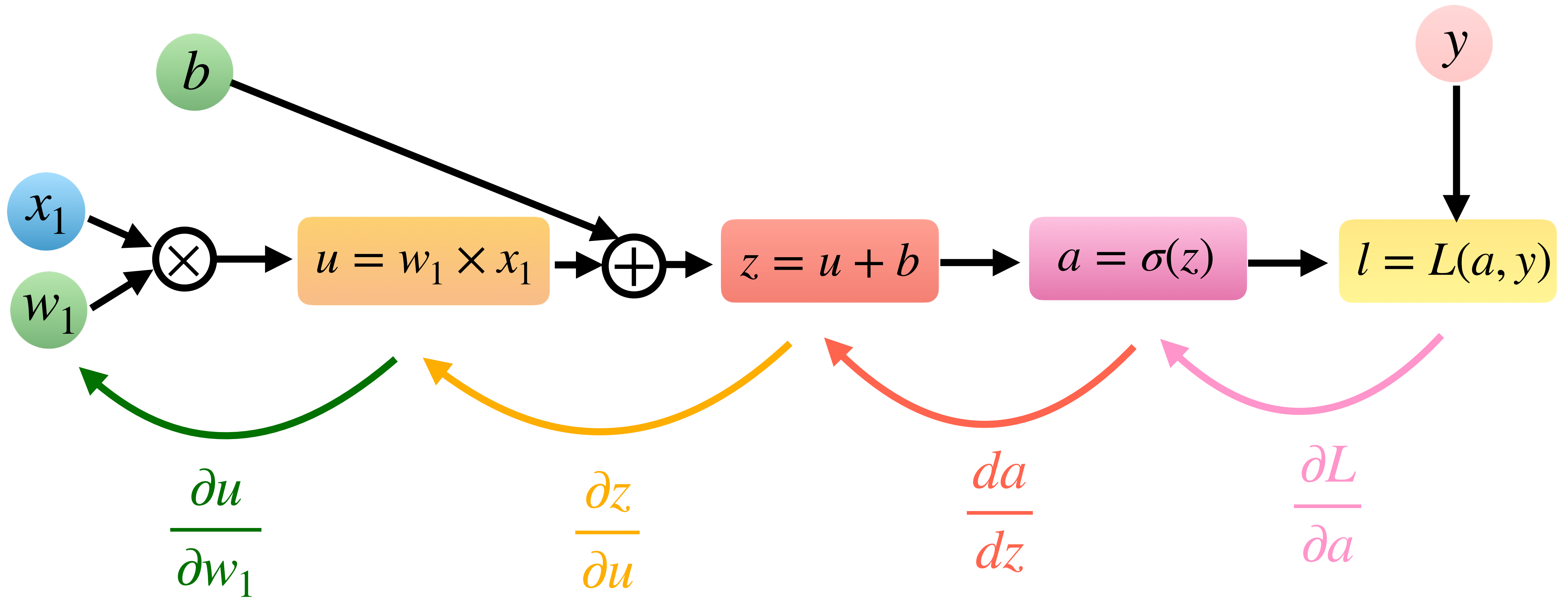
Chain Rule



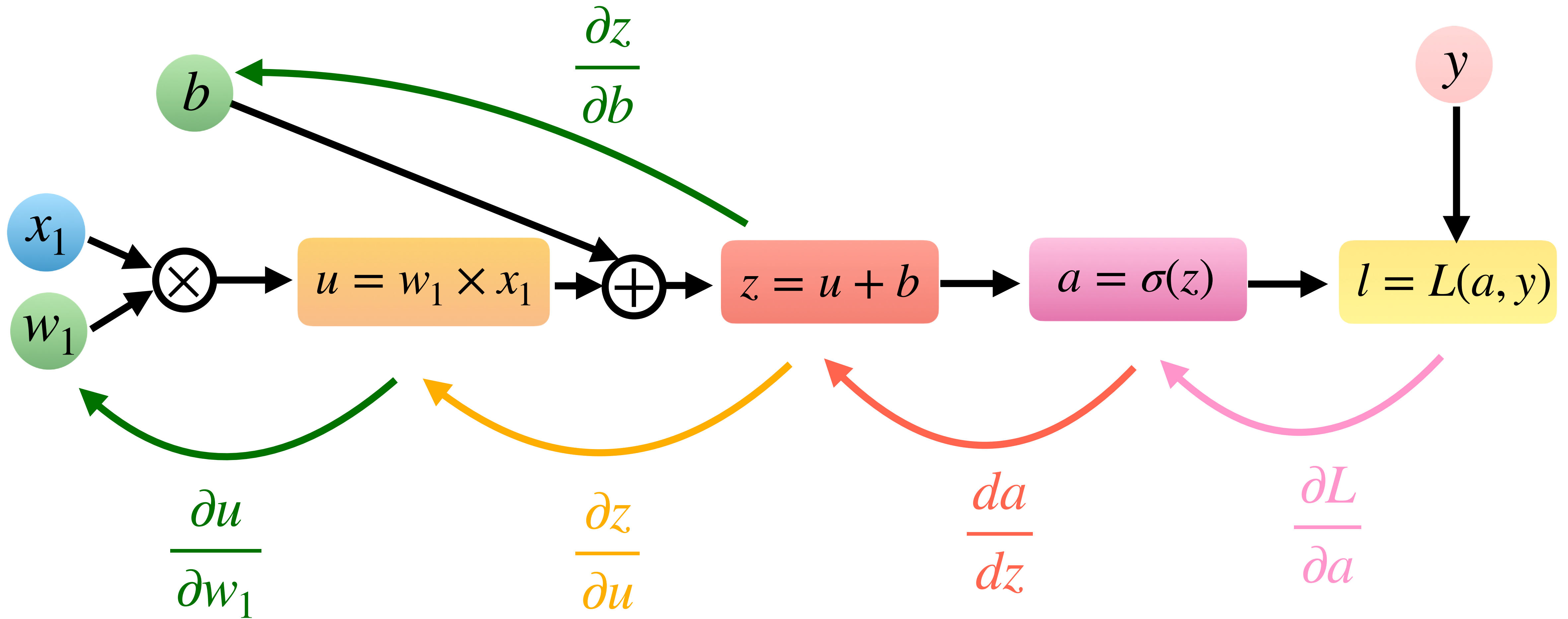
Chain Rule



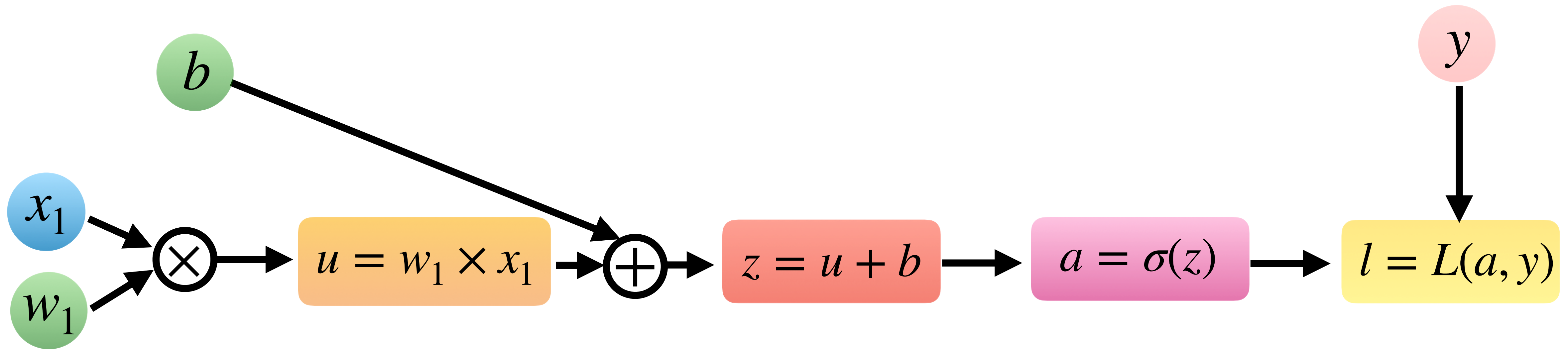
Chain Rule



Chain Rule



Chain Rule



$$\frac{\partial L}{\partial w_1} = \frac{\partial u}{\partial w_1} \times \frac{\partial z}{\partial u} \times \frac{da}{dz} \times \frac{\partial L}{\partial a}$$

Next: Derivatives vs Gradients